Optimization for Deep Learning (OPT4DL)

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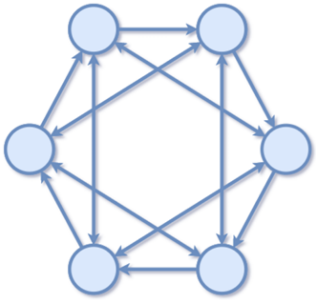
Decentralized SGD

Setup

- Assume we have K agents each with a different loss function
- Agents iterative communicate over a graph to find

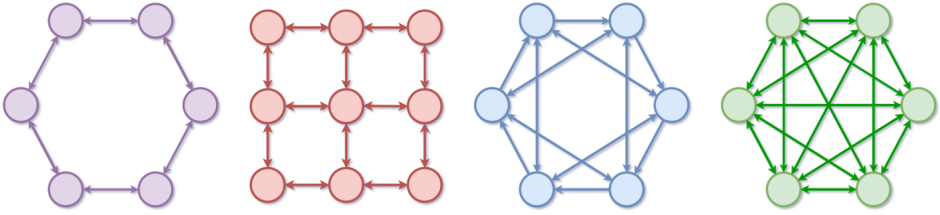
$$w^* = \arg\min_w f(w) = \frac{1}{K}\sum_{j=1}^{K} f_j(w), \qquad f_j(w) = \mathbb{E}_{z\sim D_j}[\ell(w,z)] \tag{1}$$

- At convergence, each agent's solution should be w^*, i.e., they should reach a consensus.



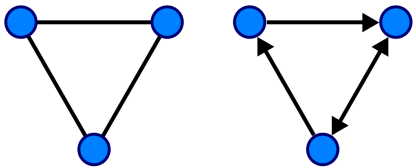
Communication Graph

- We can only hope to find a solution if the agents can communicate
- We assume a connected graph, that means there is at least one path between any two agents
- Intuitively, higher connectivity means faster convergence due to messages reaching the destination faster



Communication graph (Cont'd)

- Formally, $\mathcal{G}(\mathcal{N}, \mathcal{E})$ where $\mathcal{N} = \{1, \ldots, K\}$ is the set of agents and $\mathcal{E}$ is the set of edges
- $\{i, j\} \in \mathcal{E}$ if there is an edge from agent i to agent j, meaning, i can send a message to j
- Note, the communication could in general be asymmetric meaning $\{i, j\} \in \mathcal{E}$ but $\{i, j\} \notin \mathcal{E}$. Meaning, the graph could be a directed graph
- We will however, assume the graph is undirected for simplicity.
- Let $\mathcal{N}_i$ be the set of neighbors of agent i and note $i \in N_i$
- More generally, we assume the communication is weighted and there is a weight matrix W that governs the communication among agents such that $W \in \mathbb{R}^{K \times K}$, $0 \leq W_{ij}$



Communication graph (Cont'd)



- Since each node can send a message to itself $W_{ii} > 0$
- Since the graph is undirected $W_{ij} = W_{ji}$, meaning W is a symmetric matrix $W = W^\top$
- Since $\{i, j\} \in \mathcal{E}$ if there is an edge between agent i and agent j

$$\{i, j\}, \{j, i\} \in \mathcal{E} \qquad \equiv \qquad W_{ij} = W_{ji} \neq 0 \tag{2}$$

- For each agent j, let us also normalize the weights for incoming messages sum to 1

$$\sum_{i=1}^{K} W_{ij} = 1 \quad \equiv \quad 1^\top W = 1^\top, \qquad \text{called Column stochastic} \tag{3}$$

- And since graph is undirected and W is symmetric

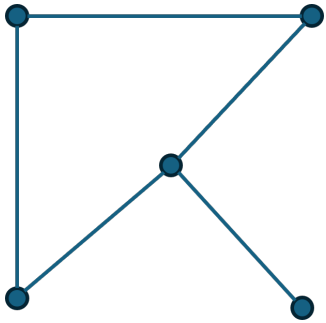
$$\sum_{j=1}^{K} W_{ij} = 1 \quad \equiv \quad W1 = 1, \qquad \text{called Row stochastic} \tag{4}$$

Doubly Stochastic Weight Matrix

- A matrix that
  is both row and column stochastic is called doubly stochastic
- Example: given any connected graph set

$$W_{ij} = \frac{1}{\max\{\text{degree}(i), \text{degree}(j)\}}, \quad i \neq j$$

  and $W_{ii} = 1 - \sum_{j \in \mathcal{N}_i} W_{ij}$
- Remark: since an agent can communicate
  with itself and the graph is connected, $\text{degree}(i) \geq 2$



Average Consensus



- Recall our goal is

$$w^* = \arg\min_w f(w) = \frac{1}{K}\sum_{j=1}^{K} f_j(w), \qquad f_j(w) = \mathbb{E}_{z\sim D_j}[\ell(w,z)] \tag{5}$$

- At convergence, each agent's solution should be w^*, i.e., they should reach a consensus.
- Let us first consider the case where $f_j(w) = 0.5\|w - x_j\|^2$, $x_j \in \mathbb{R}^d$. It tuns out f_j's and hence f are strongly convex.
- We can then easily calculate

$$w^* = \frac{1}{K}\sum_{j=1}^{K} x_j = \bar{x}, \qquad \text{Finding the average} \tag{6}$$

- This problem is called average consensus. Remember, we want to solve this over a graph
- A fully connected graph means by one exchange of local vectors x_j, each agent can compute the average (convergence in one iteration). How about general graphs?

Average Consensus Method



- Let us define a matrix $\bar{X} = \bar{x}1^\top = [\bar{x}, \ldots, \bar{x}] \in \mathbb{R}^{d \times K}$
- Then, it is not hard to see that

$$\bar{X} \cdot \frac{1}{K} 11^\top = \bar{x}1^\top \cdot \frac{1}{K} 11^\top = \bar{x}1^\top = \bar{X} \tag{7}$$

- But, for a fully connected graph, $W = \frac{1}{K} 11^\top$ (all entries are $1/K$ using a uniform weighting). That is,

$$\bar{X} = \bar{X}W, \qquad \text{a fixed point condition} \tag{8}$$

- Turns out this is true for all connected graphs with any doubly stochastic W.
- How do we find fixed points? Fixed point iteration (recall the connection to GD)
- Our method is then: Initialize local solutions x_1^i and run the following local step for t iterations

$$x_{t+1}^i = \sum_{j \in \mathcal{N}_i} W_{ij} x_t^i, \qquad \equiv \qquad X^{t+1} = X^t W \quad \text{in matrix form} \tag{9}$$

where $X^t = [x_t^1, \ldots, x_t^K] \in \mathbb{R}^{d \times K}$.

Convergence of Average Consensus Method



Theorem

Assume the graph is connected and W is doubly stochastic. Define the consensus error as

$$e_t = \frac{1}{K}\|X^t - \bar{X}\|_F^2 = \frac{1}{K}\sum_{j=1}^{K}\|x_t^j - \bar{x}\|^2 \tag{10}$$

Then, using the Banach Fixed point theorem, it holds that $e_{t+1} \leq |\lambda_2(W)|^2 e_t$, where $|\lambda_2(W)|$ is the second largest eigenvalue of W (in magnitude)

- Assuming W is doubly stochastic and recalling $W1 = 1$ means $\lambda = 1$ is an eigenvalue and in fact the maximum eigen value of W.
- The quantity $0 \leq \rho := 1 - |\lambda_2(W)| \leq 1$ is called the spectral gap of W
- Large ρ means better connectivity and faster convergence ($\rho = 1$ for a fully connected graph)
- small ρ means worse connectivity and slower convergence ($\rho = 0$ for a disconnected graph)

From consensus to Training: decentralized SGD



- Recall our problem is

$$w^* = \arg\min_w f(w) = \frac{1}{K}\sum_{j=1}^{K} f_j(w), \qquad f_j(w) = \mathbb{E}_{z\sim D_j}[\ell(w,z)] \tag{11}$$

and for consensus

$$x_{t+1}^i = \sum_{j\in\mathcal{N}_i} W_{ij} x_t^i, \qquad \equiv \qquad X^{t+1} = X^t W \quad \text{in matrix form} \tag{12}$$

- That is, everyone shares their model updates and perform local aggregation. For general training problems, we can follow a similar pattern:

$$x_{t+1}^i = \sum_{j\in\mathcal{N}_i} W_{ij}\left(x_t^i - \eta g_t^i\right), \qquad \equiv \qquad X^{t+1} = \left(X^t - \eta G^t\right) W \quad \text{in matrix form} \tag{13}$$

where $G^t = [g_t^1, \ldots, g_t^K] \in \mathbb{R}^{d\times K}$.

Convergence Analysis



Theorem

Assume the graph is connected and W is doubly stochastic with spectral gap ρ. Define the consensus error as

$$e_t = \frac{1}{K}\|X^t - \bar{X}^t\|_F^2 = \frac{1}{K}\sum_{j=1}^{K}\|x_t^j - \bar{x}_t\|^2 \tag{14}$$

where $\bar{x}_t = \frac{1}{K}\sum_{j=1}^{K} x_t^j$, and $\bar{X}^t = [\bar{x}_t, \ldots, \bar{x}_t] \in \mathbb{R}^{d\times K}$. Assume f is L-smooth. Furthermore assume we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, If $\eta = \mathcal{O}(\rho/\sqrt{T})$,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(\bar{x}_t)\|^2 + \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[e_t] = \mathcal{O}(\frac{1}{\sqrt{T}}\text{poly}(\frac{1}{\rho})). \tag{15}$$

- The average iterates $\bar{x}_t$ (an unknown quantity to the agents) is a first-order solution
- The consensus error vanishes so everyone is reaching $\bar{x}_t$ (consensus)

Proof Sketch



- Instead of studying the performance of K local models, we consider how the average model $\bar{x}_t$ (a virtual sequence unknown to the agents) performs. At the same time, we study the evolution of the important source of error, the consensus error.
- For consensus error, using our assumptions we can establish

$$\mathbb{E}[e_{t+1}] \leq \rho \mathbb{E}[e_t] + \text{ other terms} \tag{16}$$

  immediate reminding us of the potential-based analysis
- We then define a potential function

$$\Phi_t = \mathbb{E}[f(\bar{x}_t)] - f^* + C_e \mathbb{E}[e_t] \tag{17}$$

- Study the evolution of potential by summing over its consecutive differences
- When $\rho < 1$ we can find a good constant C_e to ensure convergence.